

KARMOY HAUGESUND, NorwayWMO: **014080**Lat: **59.350N**Lon: **5.198E**Elev: **26**StdP: **101.01**Time Zone: **1.00 (EUC)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.1	-6.0	-14.0	1.1	-5.6	-11.3	1.4	-3.5	17.1	4.5	15.7	3.9	3.4	100	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	23.0	16.5	20.9	15.9	19.1	15.0	17.7	21.5	16.7	19.6	15.8	18.2	4.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.5	18.9	15.3	10.9	17.9	14.6	10.4	17.2	49.9	21.6	46.9	19.5	44.3	18.3	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.7	13.9	12.3	-10.3	27.0	2.6	2.2	-12.2	28.6	-13.7	29.9	-15.2	31.1	-17.1	32.7		
			-10.8	19.1	2.6	1.2	-12.7	20.0	-14.2	20.7	-15.7	21.4	-17.6	22.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	2.0	2.1	3.6	6.6	9.8	12.4	15.1	15.2	13.0	9.0	5.7	3.5
	DBStd	5.55	3.62	3.32	2.66	2.75	3.12	2.29	2.65	2.03	2.27	2.62	3.34	3.87
	HDD10.0	1200	247	222	199	109	40	3	0	0	3	48	130	200
	HDD18.3	3711	505	455	458	352	265	179	109	102	162	288	378	458
	CDD10.0	545	0	0	0	6	34	74	158	160	92	18	2	0
	CDD18.3	14	0	0	0	0	1	1	9	3	1	0	0	0
	CDH23.3	62	0	0	0	0	4	5	43	8	1	0	0	0
	CDH26.7	9	0	0	0	0	0	0	8	1	0	0	0	0
	WSAvg	6.0	7.2	6.2	5.8	5.3	5.6	5.4	5.0	5.5	6.1	6.7	6.6	7.2
(15) Precipitation	PrecAvg	1593	162	130	117	90	76	85	97	144	159	183	175	190
	PrecMax	1928	297	248	230	155	149	154	165	258	269	283	320	315
	PrecMin	1069	11	28	2	41	25	24	46	36	45	61	35	39
	PrecStd	230	75	65	55	26	30	35	31	48	67	65	76	72
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	8.3	10.1	18.2	23.5	23.1	28.0	24.3	21.7	16.7	13.3	10.4
		MCWB	7.5	6.9	6.5	10.6	14.6	14.7	18.6	17.1	16.3	13.7	11.0	9.3
	2%	DB	8.0	7.5	8.2	15.0	19.8	19.7	24.0	21.4	19.0	14.6	11.7	9.7
		MCWB	7.0	6.5	6.3	9.0	13.0	14.3	17.5	16.1	14.1	12.2	10.2	8.7
	5%	DB	7.4	7.0	7.4	12.7	16.7	17.6	21.3	19.7	17.3	13.5	10.8	9.0
		MCWB	6.4	6.2	5.9	8.2	12.3	13.7	16.8	15.6	14.1	11.4	9.2	8.1
	10%	DB	6.8	6.4	6.9	10.6	14.7	15.8	19.2	18.2	16.3	12.7	10.0	8.2
		MCWB	5.8	5.5	5.6	7.2	10.8	12.8	16.0	15.0	13.7	10.7	8.6	7.0
	0.4%	WB	8.0	7.4	8.0	11.2	16.3	16.4	19.6	18.4	17.5	14.2	11.6	9.7
		MCDB	8.1	7.9	8.8	16.3	22.6	19.9	25.2	21.7	20.6	15.7	12.3	9.9
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	7.4	6.8	7.0	9.7	14.0	15.0	18.2	17.0	15.7	13.1	10.5	9.0
		MCDB	7.7	7.2	7.5	13.2	18.0	18.4	22.6	19.5	17.4	14.1	11.5	9.5
	5%	WB	6.7	6.3	6.4	8.8	12.6	14.0	17.3	16.2	14.9	12.2	9.8	8.3
		MCDB	7.3	6.9	7.1	11.5	16.2	17.0	20.8	18.5	16.7	13.2	10.5	8.9
	10%	WB	5.8	5.6	5.9	7.9	11.3	13.1	16.2	15.5	14.1	11.1	8.9	7.2
		MCDB	6.5	6.4	6.6	9.9	14.0	15.3	18.9	17.8	15.8	12.4	9.7	8.1
	5% DB	MDBR	4.2	4.4	4.5	5.9	5.8	5.1	5.0	5.2	5.1	5.0	4.5	4.2
		MCDBR	4.1	3.5	4.5	9.2	10.0	8.3	9.3	7.3	6.6	5.6	3.9	3.8
		MCWBR	4.0	3.4	3.4	5.1	5.2	4.3	4.6	3.8	4.0	4.3	3.8	4.2
		MCWBR	3.9	3.4	3.7	7.2	9.0	7.1	7.8	5.7	5.5	4.2	4.0	3.8
(40) Clear-Sky Solar Irradiance	taub		0.308	0.333	0.352	0.381	0.368	0.363	0.377	0.376	0.360	0.339	0.311	0.286
		taud	2.167	2.278	2.338	2.309	2.420	2.466	2.445	2.444	2.474	2.410	2.241	2.105
	Ebn at Noon		467	670	780	819	863	872	848	820	772	664	466	348
		Edh at Noon	47	70	88	108	103	100	100	94	78	61	43	34
(44) All-Sky Solar Radiation	RadAvg		0.27	0.84	2.03	3.74	5.17	5.53	4.84	3.73	2.37	1.15	0.39	0.17
	RadStd		0.04	0.14	0.27	0.34	0.56	0.65	0.46	0.34	0.21	0.14	0.05	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-117	N/A	N/A

Nomenclature: See separate page